

Auditable digital-oracle resources for controlled compound Asian-basket pricing

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Abstract

Quantum mean-estimation guarantees do not by themselves establish a practical option-pricing advantage. We study a controlled compound Asian-basket pricing implementation with an explicit finite input law, reversible source, classical comparison and finite-confidence estimation schedule. Sound integer ranges allow narrower exact multipliers; parallel schedules reduce the arithmetic depth while paying for extra workspace. A concrete Hadamard-test implementation and a bounded-output amplitude estimator substantially improve the earlier quantum implementation. Their costs still do not establish the declared classical crossover. We separate verified digital computation from remaining financial certification and conditional physical-resource assumptions.

Keywords: quantum algorithms; option pricing; amplitude estimation; reversible arithmetic; resource estimation; classical control variates.

1. Question and contribution

The question is whether a compiled quantum estimator can price a financially specified compound Asian-basket option with a meaningful complete-latency advantage over strong controlled classical methods. The acceptance target is tenfold latency improvement at one-cent absolute price error and 99% per-price confidence, under a declared physical-qubit cap of ten million. Tenfold is a practical robustness margin for uncertain future execution assumptions; it is not a statistical significance threshold. A successful selected instance would still require confirmation over a frozen parameter region. No such confirmation has been performed because the prerequisite gates do not pass.

We contribute an auditable case study of a particular implementation. It joins exact digital source execution, explicit estimation constants, a range-dependent compiler optimization, a finite-law control-bias analysis, and strong classical comparators. The range and schedule transformations retain the digital output; their improvements are relative to a named earlier quantum implementation. They are not improvements over classical pricing.

The negative crossover outcome is scoped. A constructed schedule that is slow does not lower-bound every possible circuit. A capacity inequality applies only to the fixed operation workload and throughput model used to derive it. Neither result proves quantum advantage impossible for these options.

2. Relation to existing work

Blanchet et al. [1] explicitly discuss compound-option pricing as nonlinear Monte Carlo. Consequently, neither selecting a compound payoff nor introducing nested quantum estimation establishes originality. Kothari and O'Donnell [2] already provide variance-sensitive source-code mean estimation and describe

Hadamard tests as an alternative to phase estimation. Our contribution is a concrete finite-confidence realization and its cost for this source, not a new mean-estimation theorem. The bounded alternative uses standard amplitude estimation with the explicit guarantee summarized by Montanaro [3].

Chakrabarti et al. [4] and Wang and Kan [5] already perform detailed derivative resource estimates including state preparation and arithmetic. Herman et al. [6] formulate a derivative-pricing framework that separates truncation, discretization, arithmetic and estimation errors. Our particular midpoint Box-Muller law, guarded paths and control combination require their own quantitative analysis; we do not claim a first general error framework. Lemaire, Montes and Pages [7, Section 4.1] explicitly account for bias from a quantized approximate control expectation and price a compound option using analytic inner integration [7, Section 4.2.2]. Approximate-control bias and removal of nesting by conditional formulas are therefore established principles.

The proposed journal contribution is the reusable, inspectable integration and its specific numerical findings. This bounded prior-art comparison is not a proof of publication priority or sufficient journal novelty.

3. Financial contract and classical comparison

We use equal-weight arithmetic Asian baskets under correlated geometric Brownian motion, spot 100, risk-free rate 0.03 and inner strike 100. The inner payoff averages all specified monitoring dates and assets. A compound call at time $\tau=T/2$ pays $\max(C(X)-K_c, 0)$, where $C(X)$ is the time- τ continuation price of the inner call conditional on the past state X . Its time-zero price is $\exp(-r \cdot \tau) E[\max(C(X) - K_c, 0)]$. Monitoring is discrete by financial definition; the Gaussian probability law is continuous. A finite midpoint law used by the circuit is a further numerical approximation, not a redefinition of the target.

Model	Assets	Dates	Volatility	Correlation	Maturity
C4	4	12	0.20	0.20	1
C8	8	12	0.40	0.20	1
H4	4	12	0.40	0.70	3
H8	8	24	0.40	0.20	3

Development uses compound strikes 3, 6 and 9. The detailed emitted quantum source comparison uses C4/H8 and strike 6. C4/H8 and all earlier development cases are not independent confirmation. In particular, D1/D2 belong to the separate original manuscript and do not validate the present model.

The classical pipeline exploits a conditional linear/geometric control, moment-matched approximation, regression where useful, conditional sampling, randomized QMC, and antithetic correction diagnostics. It is not plain Monte Carlo. The archived favorable screening times are 11.1059177 seconds for C4 and 13.847206 seconds for H8, including all three strikes. Charging that full three-output time against one quantum strike deliberately favors quantum in this rejection screen. H8 uses the faster preserved controlled cap/tail result; a later slower parity run does not replace it just to improve the quantum ratio. These empirical randomized-QMC intervals and reference-overlap diagnostics are not distribution-free rigorous confidence certificates.

A separate fixed-iid C4 comparator costs 267.8373668 seconds including its archived training charge, under the stated iid/exact-evaluation premises. It provides a second, more conservative confidence-contract screen. Neither comparator supplies a completed certificate for the quantum implemented policy. Both sides must receive the same useful controls and comparable opportunities for preprocessing and amortization in any future affirmative comparison.

4. Controlled decomposition and error obligations

Let $a(X)$ be the already accrued arithmetic average contribution and let A and G be the future arithmetic and geometric averages. Write $k=2(100-a)$, $q=\exp(-r(T-\tau))/2$, $F=q(E[A|X]-k)$, and $P_g=q E[(k-G)+|X]$. Define

$$\begin{aligned} c_0 &= F + P_g, \\ R &= q*((k-G)+ - (k-A)+). \end{aligned}$$

Under the intended continuous model $C=c_0-E[R|X]$. For a computable approximation $V(X)$, take the decision $d(X)=1\{V(X)>K_c\}$ and the signed residual $Y=d(X)*(c_0-V-R)$. Then

$$E[(V-K_c)+] + E[Y] = E[d(X)*(C(X)-K_c)].$$

The optimal compound value adds nonnegative regret $E[(C-K_c)+-d(C-K_c)]$. This is the tower property and elementary algebra. It is useful because the baseline and residual should be analyzed jointly: their V terms cancel. It does not make implemented-policy regret free. A circuit uses a clipped finite-law version of Y , so transfer of the continuous analytic control identity must also be bounded rather than assumed exact.

The price budget remains 0.002 for numerical errors, 0.003 for the baseline expectation, 0.002 for the residual mean and 0.003 for policy regret. The failure budget is 0.002 baseline, 0.001 moment acquisition, 0.001 regret, 0.003 ideal estimation, 0.0005 coherent phase-function error, 0.0005 rotation synthesis and 0.002 physical execution. Bounded-output QAE does not need moment acquisition or the phase-function oracle; unused allowances are not silently reallocated.

The previously certified $q=32$ law/clipping/control bridge is below 0.000029 for the stated models in real arithmetic. This follow-up proves complete integer ranges for $f=40/q=32$ and bounds the joint arithmetic expectation by 0.000099876, 0.000051936, 0.000011763 and 0.000021978 dollars for C4, C8, H4 and H8 respectively. Independent internal review found no remaining blocker for this restricted claim: baseline and correction must use exactly the same digital policy and decision. The finite-policy regret, its baseline-confidence transfer and the tight digital second moment remain distinct obligations. No complete one-cent/99% continuous-model quantum price is reported.

5. Digital source and verified compiler changes

Each random input is an independent $q=32$ midpoint-uniform variate prepared by Hadamards. Reversible Box-Muller generation supplies correlated normals without QRAM or an exponentially large Gaussian table. The financial source uses 40 fractional bits in 72-bit signed words, guarded spots between 2^{-16} and 4096, and the archived log-state reset at the conditioning time. Elementary function coefficient tables, arithmetic floors and the reset are included in the model bridge. All source temporaries are retained; this deliberately simple reversible storage policy is not an optimized memory lower bound.

Sound interval propagation over the entire reachable finite support initially loses correlations in repeated variables. Structural witnesses for normalized logarithms, interpolation coordinates, exponent range reduction and moment ratios recover the needed dependencies. The full financial graphs of all four models have no unresolved signed arithmetic overflow sites. Canonical expression binding transfers these intervals to the pruned, common-subexpression-eliminated source without assuming graph identity from a few test cases.

For a multiplication with certified signed operand widths a and b , we retain $n = \min(w+f, a+b)$ product bits. Unsigned partial products plus the two signed correction terms compute the signed product modulo 2^n ; the double-sign term vanishes modulo 2^n . The output is the f -shifted slice, sign-extended when the entire product is shorter than the output. The clean XOR circuit is therefore identical to the original fixed-point multiplication on the certified domain. Outside that domain it remains reversible but need not compute the same value.

Parallel source schedules group independent leaves by dependency level and allocate disjoint input/scratch slots to every simultaneously active leaf. Copies from a shared SSA control are charged in separate rounds. Inverse waves repeat the clean XOR leaves in reverse dependency order. This is a valid coherent inverse on their clean-workspace subspace. It does not assume a serial-depth count is an optimal schedule or free shared workspace.

Source	New T count	New wave T depth	Source logical qubits	T improvement	Depth improvement
C4 f40	568,839,250	11,609,892	855,344	2.3684x	38.4312x
H8 f40	2,089,294,634	15,258,820	3,153,512	2.3894x	108.1126x

Improvements use the preceding exact truncated-multiplier serial source as the denominator. Whole-source basis execution checks the emitted primitive gates, nonzero output XOR, preserved inputs and cleanup against exact integer targets. Exhaustive small-width tests and production edge cases supplement the proofs. They are not executions of the complete quantum estimator.

6. Explicit estimators

6.1 Variance-sensitive Hadamard schedule

Suppose $E[Y^2] \leq s^2$ and the current mean interval is $[a, a+W]$. For rational $0 < c < 1$ set $e = W/(1+3c)$, $t = a + c \cdot e$, $z = (Y - t)/N$ and $\epsilon = e/N$, with N a power of two. Since $|t| \leq s$, $\sqrt{E[z^2]} \leq s_0 = 2s/N$. Theorem 3.21 of [2] bounds the eigenphase of $U = (2|\psi\rangle\langle\psi| - I)\text{diag}(\exp(-2i \operatorname{atan}(z)))$ on a spectral event of probability at least $1 - 2/C^2$, for $C \cdot s_0 < 1$. Writing $m = |E[z]|$, safe bounds are

$$\left| \frac{2m}{(\sqrt{1+s_0^2})(1+C \cdot s_0)} \right| \leq |\theta| \leq 2 \arcsin(m/(1-C \cdot s_0)).$$

The maintained interval also gives $m \leq (1+2c) \cdot \epsilon$. One Hadamard test with controlled U^T has plus probability $E[(1 + \cos(T \cdot \theta))/2]$. Interval-certified trigonometric bounds distinguish $m \leq c \cdot \epsilon$ from $m \geq \epsilon$. We select an integer T and an integer count threshold; exact rational binomial tails bound both errors. Stage allocations sum to 0.003. If the test says small, retain $[a, a+qW]$, and otherwise $[a+2cW/(1+3c), a+W]$, where $q = (1+c)/(1+3c)$. Both updates remain valid for permitted answers in the ambiguity gap. Exact rational controller intervals prevent accumulated endpoint drift.

The selected C4/H8 tight-moment designs use 1,203,320/2,967,610 controlled iterations, versus 1,540,372,419/6,287,324,746 in the earlier explicit QPE implementation. This 1,280x/2,119x reduction is a quantum implementation result. Its tight second moments are still conditional on digital transfer. The phase source is $f=64$; actual previously certified Clifford+T rotation strings are reused, with their new multiplicities paid.

6.2 Unconditional digital bounded-output schedule

The digital residual obeys $|Y| < 100$. Add 48 uniform selector bits u and compute $u \leftarrow \text{raw}(Y) + 128 \cdot 2^{40}$. The exact flag probability is $p = E[(Y+128)/256]$. The shift is an exact high-bit flip in the relevant signed word, followed by a clean comparator. No data-dependent rotation or second-moment certificate is needed. The controlled Grover iterate computes financial source, selector, predicate phase, selector inverse and financial inverse, then reflects on all random and selector bits with the controlled relative sign retained.

The standard QAE bound is $2\pi \sqrt{p(1-p)} / (M + \pi^2 / M^2)$ with success at least $8/\pi^2$ [3, Theorem 2]. Uniformly bounding $\sqrt{p(1-p)}$ by $1/2$ gives $M=524,288$ for the required mean error. Fifteen repetitions and a median bound failure below 0.003 by an exact binomial calculation. The total is 7,864,305 controlled iterations. State preparation, both inverse passes, reflections, IQFT and synthesized rotations are included. This establishes a digital residual mean estimator, not the full continuous-model price.

7. Combined costs and conditional capacity

The following results include the new source, explicit estimator and actual synthesized rotations. Seconds are a deliberately favorable sensitivity at one nanosecond per scheduled T layer with Clifford time and physical overhead set to zero. They are not quantum hardware timings.

Method	Model	Total logical T	Logical qubits	Scheduled seconds at 1 ns/T layer	10x budget, seconds
Hadamard, conditional tight moment	C4	6.9976e14	879,289	19,399.5	1.11059
Hadamard, conditional tight moment	H8	6.2381e15	3,182,449	58,849.2	1.38472
Bounded QAE, unconditional digital support	C4	4.4738e15	857,599	91,501.7	1.11059
Bounded QAE, unconditional digital support	H8	1.6432e16	3,160,759	120,669.1	1.38472

Independent tests could run on multiple source copies within an adaptive stage. Their full workspaces must then be paid. The table is a constructed single-copy schedule, and its failure does not preclude all alternative scheduling choices.

An orthogonal necessary capacity condition is $t \geq N_T \cdot t_{\text{lane}} \cdot q_{\text{lane}} / P$ for P available physical qubits, q_{lane} physical qubits per T-consuming lane, and lane interval t_{lane} . It deliberately gives all qubits to consumption lanes and makes source data, routing and factories free. With $P=10^7$, $q_{\text{lane}}=1$ and $t_{\text{lane}}=1$ ns, the arithmetic-only Hadamard floors are 0.06997/0.62379 seconds. Both pass this

extremely permissive necessary screen; claiming otherwise would overstate the evidence. At 17 physical qubits per lane they are 1.18946/10.60439 seconds. These are assumptions about a fixed implementation, not a theorem about arbitrary quantum computers or native non-Clifford gates.

For the stated 15-to-1 factory family, even a final-stage-only capacity bound with all ten million qubits assigned to factories, distance 3 and a one-ns code cycle gives 431.77/3,849.39 seconds for the conditional Hadamard arithmetic. Lower factory stages, rejection, data storage, routing and synthesis would add cost. This fails both empirical screening budgets and the fixed-iid C4 budget of 26.7837 seconds. The actual parallel source allocation also exceeds the ten-million cap under a distance-3 patch assignment. Other factories, native resources, storage policies and device models are not ruled out by this bound.

The complete crossover equation is

$$T_{\text{setup_Q/A}} + T_{\text{baseline}} + T_{\text{regret}} + T_{\text{mean}} + T_{\text{physical_IO}} \leq T_{\text{classical}}/10,$$

where A is the declared amortization count. None of these terms may be silently removed to claim a full price. Positive setup/certification costs strengthen the failure of the studied mapping. Repeated prices do not amortize the coherent iterations that must be performed for each required output.

8. Reproducibility, limits and conclusion

The repository retains prior evidence and adds separate source, coefficient, range, estimator, schedule and capacity artifacts. Hashes bind emitted leaves to manifests; an artifact manifest binds this release. The pinned research environment and reproduction commands are provided with the companion report. Separate AI agents review work they did not author; this is an internal check, not journal peer review or a replacement for named human authors' responsibility.

The financial range proof and digital-estimator guarantees do not imply a completed continuous-price certificate. Policy regret, baseline confidence and tight-moment transfer must retain their explicit status. No device layout or decoder was validated, no full QAE price was simulated or run on hardware, and no held-out advantage confirmation was attempted. The possible benchmark contribution requires human mathematical review and an editorial assessment of novelty before submission.

No defensible significant quantum advantage established yet. The work removes substantial implementation weaknesses without closing the practical crossover. The result is useful evidence about this compiled workload; it does not fulfill the original advantage objective or establish an impossibility result. A future reopening needs a quantified change in complete execution cost, not a relabeling of the implementation improvements reported here.

Data, code and assistance statements

Source and generated evidence are in the accompanying repository under `research/controlled_priority_completion`, `results/controlled_priority_completion` and `docs/controlled_priority_completion`. The release manifest specifies the exact artifacts. A permanent archival DOI and an author-approved data statement remain to be supplied before submission.

AI coding assistants were used extensively in investigation, implementation, derivation, internal review and drafting. Named human authors have not yet approved this draft or certified their responsibility for it. Author contributions, funding, competing interests, originality and concurrent-submission declarations must be supplied and approved by those authors; none is inferred here.

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